

BREWTON 3 NNE, AL, USA

WMO: 722277

Lat: **31.1449N**Lon: **87.0518W**Elev: **170**StdP: **14.61**Time Zone: **-6.00 (NAC)**Period: **08-19**WBAN: **63899****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.8	28.2	N/A	N/A	N/A	N/A	N/A	N/A	13.5	48.2	12.4	48.6	3.7	10	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	93.3	N/A	91.7	N/A	90.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.6	8.9	7.9	DB	19.4	95.4	3.9	2.4	16.6	97.2	14.4	98.6	12.2	100.0	9.4	101.7
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.8	48.5	53.5	59.0	65.6	72.5	78.8	79.9	79.6	76.5	66.7	56.1	52.6
	DBStd	13.17	10.39	9.60	8.50	5.98	5.21	3.05	2.54	2.75	4.23	7.53	8.38	9.72
	HDD50	373	157	69	23	1	0	0	0	0	0	1	33	89
	HDD65	1910	519	335	221	65	12	0	0	0	1	72	287	398
	CDD50	6154	109	167	301	468	699	865	927	917	794	519	218	170
	CDD65	2213	6	14	34	82	245	415	462	452	344	125	21	13
	CDH74	18743	11	51	240	748	2176	3805	4095	3757	2696	1019	112	33
	CDH80	7310	0	0	23	119	761	1703	1784	1584	1062	264	10	0
Wind	WSAvg	3.0	3.8	4.0	4.1	3.6	2.7	2.5	2.2	2.2	2.2	2.7	2.8	3.5
Precipitation	PrecAvg	63.7	5.2	5.1	6.5	4.6	4.9	6.3	7.2	6.1	5.4	3.4	4.9	4.5
	PrecMax	95.4	12.6	10.2	15.8	11.1	16.0	15.5	19.1	14.7	25.2	15.0	13.0	12.5
	PrecMin	43.3	1.2	1.2	0.8	0.5	0.2	1.1	1.7	1.7	0.9	0.0	1.2	0.9
	PrecStd	10.5	2.2	2.3	3.7	2.9	3.2	3.2	3.1	3.1	4.6	3.1	2.7	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.7	78.5	82.7	85.6	91.9	96.0	94.6	94.9	93.2	89.0	81.2	77.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	72.4	75.5	79.8	83.3	89.1	93.4	92.4	92.3	90.6	85.7	77.4	74.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	69.1	72.1	76.9	81.2	86.7	91.3	90.8	90.5	88.7	83.0	74.3	71.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	65.7	69.2	73.4	78.8	84.7	89.1	88.9	88.4	86.5	80.1	71.3	68.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	22.3	22.2	23.5	23.4	21.9	19.4	18.2	17.7	19.3	23.3	24.4	20.0
		MCDBR	21.7	21.8	23.8	25.3	24.4	22.6	20.7	20.6	21.7	24.4	24.0	19.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.326	0.343	0.365	0.393	0.436	0.466	0.525	0.508	0.442	0.378	0.349	0.332	
	taud	2.540	2.494	2.426	2.351	2.264	2.236	2.085	2.129	2.289	2.445	2.470	2.532	
	Ebn at Noon	287	292	291	285	272	263	247	250	263	275	275	279	
	Edh at Noon	27	31	36	40	43	45	52	49	40	32	29	26	
All-Sky Solar Radiation	RadAvg	938	1120	1515	1847	2033	1956	1873	1749	1607	1399	1067	819	
	RadStd	68	147	128	152	122	159	120	116	107	154	95	82	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (40 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+175

Nomenclature: See separate page